

MMLDM: Multimodal Latent Diffusion Model for Text-to-Time Series

Spectral Dual-Latent VAE + DiT Flow Matching + Text-Guided Feature Modeling

VAE Encoder

VAE Decoder

Latent Space

DiT Model

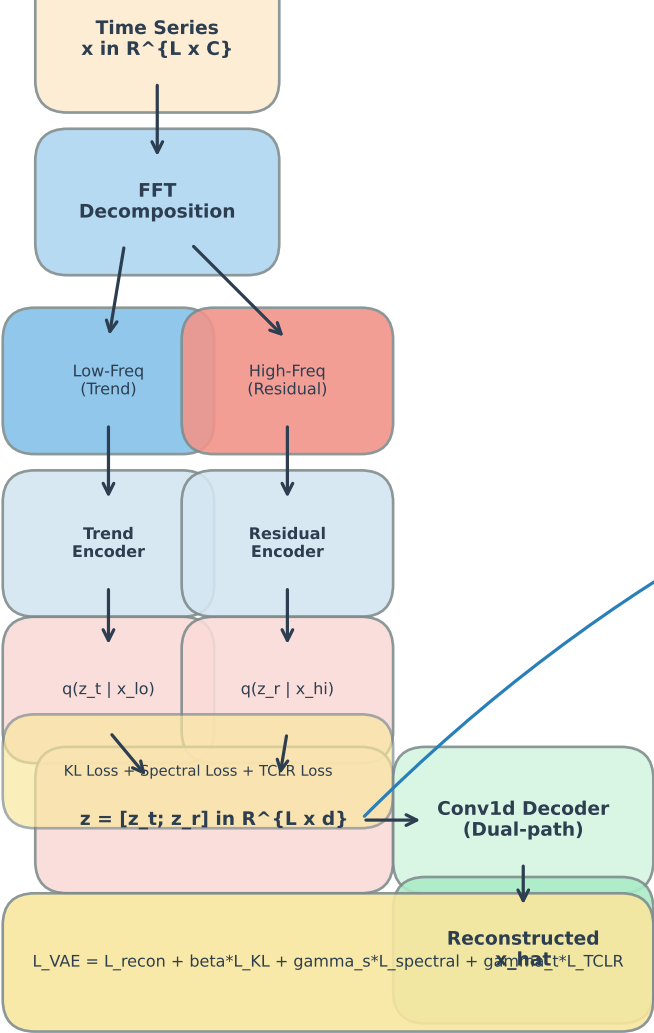
Text Processing

Innovations

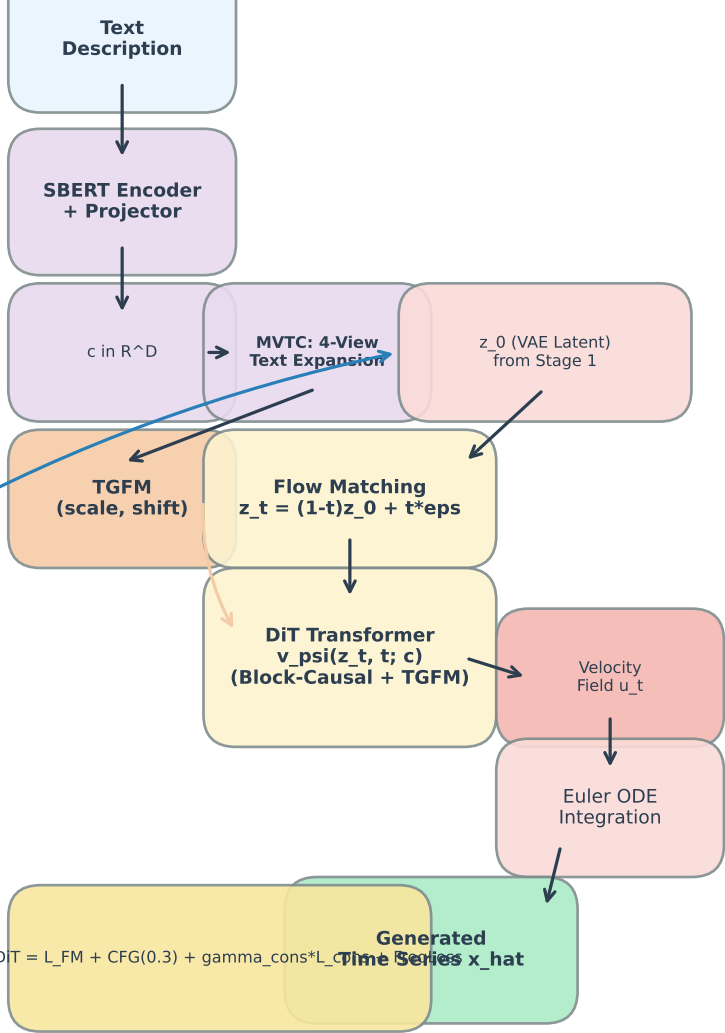
Loss Terms

Output

Stage 1: Spectral Dual-Latent VAE



Stage 2: Multimodal DiT with Flow Matching



[V2] feature/mmv2-spectral-dual-latent

[Verb] myverbal

[V4] mmlm-v4

Full V2 pipeline: Stage 1+2, all innovations A-F

V2 + Weather dataset (21-var) + CTFP/PID/PTSD evaluation

V2 codebase + warmup fix (clean training baseline)

Inference Pipeline

